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CN Assignment # 02 SecSA

Task #01

1) Reliable data transfer is important Because the network carries critical information such as casualty lists, supply requests and rescue assignment. If a packet is lost, duplicated or corrupted it could cause incorrect decision and can put life on danger Therefore reliable data transfer is important.

2) Stop and Wait can be inefficient because the link has a high RTT of 500 ms. The sender will have to wait for Ack before sending next packet. Link will be idle

$$\text{Transmission} = \frac{8000}{2 \times 10^6} = 4\text{ms}$$

→ Transmission rate is less so more time is spent on waiting.

3) Because pipelining ~~allows~~ allow to send multiple packets without Ack improving link utilization. During the log satellite RTT time this protocol can send multiple packets. That increases throughput and allows info to reach rescue team more efficiently.

4) Selective Repeat (SR) is more suitable for this scenario because firstly ① SR retransmits only lost or corrupted packets whereas GBN resends all. Secondly ② SR allows to buffer out of order packets which reduces unnecessary retransmission over high RTT link.

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Task # 02

a) Go - Back - N (GBN)

In GBN out of order packets are discarded. receiver also expected next packets.

Event	Sender packet	Receiver behaviour	ACK	Sender window
1	Send Pkt 0	Receives Pkt 0	Ack1	[0-3]
2	Send Pkt 1	Receives Pkt 1	Ack2	[0-3]
3	Send Pkt 2	Packet 2 lost	None	[0-3]
4	Send Pkt 3	Discard 3 (out of order)	Duplicate Ack2	[0-3]
5	Send Pkt 4	" 4 (out of order)	Duplicate Ack2	[1-4]
6	Send Pkt 5	" 5 (out of order)	Duplicate Ack2	[2-5]
7	Receive Ack0 & 1	-	Acks 1 & 2 received	[2-5]
8	Timeout for 2	-	No new Ack	[2-5]
9	Retransmit 2	Receives Pkt 2 in order	Ack3	[2-5]
10	" 3 pkt	Receives Pkt 3	Ack 4	[3-6]
11	" 4 Pkt	Receives Pkt 4	Ack5	[4-7]
12	" 5 Pkt	Receives Pkt 5	Ack6	[5-8]
13	Send pkt 6	Receive Pkt 6	Ack 7	[6-8]
14	" pkt 7	" Pkt 7	Ack 8	[6-8]
15	" Pkt 8	" Pkt 8	Ack9	[6-1]
16	" pkt 8	" Pkt 8	Ack 1.	All acknowledged

→ Because Pkt 2 lost GBN retransmit 3,4,5 until timeout window remain [2-5]

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(b) SR → ~~Selected~~ Selective Repeat

Event	Sender Behaviour	Receiver behaviour	Ack	Sender Window
1	Send Pkt 0	Receives Pkt 0	Ack 0	[0-3]
2	" Pkt 1	" Pkt 1	Ack 1	[0-3]
3	~~Pkt 2~~ Pkt 2	Packet 2 is lost	Naive	[0-3]
4	Send Buffer Pkt 3	Buffer Pkt 3	Ack 3	[0-3]
5	Receive Ack 0	–	Ack 0 received	[1-4]
6	Send Buffer Pkt 4	Buffer Pkt 4	Ack 4	[1-4]
7	Receive Ack 1	–	Ack 1 received	[2-5]
8	Send Buffer Pkt 5	Buffer Pkt 5	Ack 5	[2-5]
9	Receive Ack 3, 4 & 5	Packet remained buffer	Individual Acks received	[2-5]
10	Timeout for Pkt 2	–	No Ack 2 yet	[2-5]
11	Retransmit Pkt 2	Receive Pkt 2 and deliver in order	Ack 2	[2-5]
12	Receive Ack 2	All 2-5 Pkt acknowledged	Ack 2 received	[6-9]
13	Send Pkt 6	Receive Pkt 6	Ack 6	[6-9]
14	" Pkt 7	" Pkt 7	Ack 7	[6-9]
15	" Pkt 8	" Pkt 8	Ack 8	[6-1]
16	" Pkt 9	" Pkt 9	Ack 9	[6-1]
17	Receive remaining Acks	All delivered & acknowledged	Ack 6-9 to	Transmission complete

→ Conclusion:

→ In conclusion GBN:- 4 packets (2, 3, 4 & 5) are retransmitted because in GBN the receiver discard out of order packets.

→ SR → In SR 1 packet is retransmitted → Pkt 2 because receiver buffers out of order packets & sends acknowledgements separately.

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Task #05

(1)

→ Pipelining improves link utilization because the sender can send multiple packets without waiting for Ack after every packet. In stop-wait the sender spends most of its waiting for Ack because of long RTT. With window of 4 the sender can send four packets in a row. good use of bandwidth.

→ Transmission rate = $\frac{8000}{2 \times 10^6}$ = 0.004s

→ Stop & wait

$U = \frac{L/R}{RTT + L/R}$ = $\frac{0.004}{0.5 + 0.004}$ = 0.00794

RTT = 500ms = 0.5s

= 0.794% → utilization

→ N = 4

$U = \frac{N(L/R)}{RTT + L/R}$ $U = \frac{4(0.004)}{0.5 + 0.004}$ = 0.03175

Utilization = 3.171% 3.171%

→ Therefore pipeling with window of 4 increases utilization from 0.794% to 3.171% : → four times higher Utilization still low because RTT is higher.

(2)

→ **Sequence Number (1)**

with 3-bit sequence number. total number of possible sequence is

$2^3 = 8$

0, 1, 2, 3, 4, 5, 6, 7

(b)

For selective repeat the maximum safe window size is half the sequence numbers

$$W_{max} = \frac{2^k}{2}$$

$k = 3$

$$W_{max} = \frac{2^3}{2} = \frac{8}{2} = 4$$

→ window should not exceed 4 packets

(c) A larger window can cause ambiguity because sequence numbers are reused after they wrap around. If the sender window is too large an old packet that arrives late may have the sequence number of a new packet. The receiver may then mistake the old packets for a new one causing incorrect data delivery.

Example

→ 0 expired

0 1 2 3 4

0-4 → packets sent 0 is delayed

5 6 7 0 1

New packets sent 5 6 7 0 → accepted

→ Problem: If the delayed old packet 0 arrives while receiver window also accepts new packet 0, the receiver may confuse the old packet with new